

K S SUDHAKAR

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SUMMARY

Computer Science graduate with a strong foundation in problem solving and a passion for building scalable software solutions. Experienced in developing applications across web and systems domains with a keen interest in leveraging emerging technologies to solve real world challenges. Proficient in Python, Java and C++, with a focus on writing clean, optimized, and maintainable code. Adept at quickly learning new tools and technologies, and driven to contribute to innovative, high impact projects in a collaborative environment.

EDUCATION

Amrita Vishwa Vidyapeetham <i>Bachelor of Technology in Computer Science and Engineering; CGPA: 7.00</i>	Coimbatore, India 2022 – 2026
SBOA Matriculation Higher Secondary School <i>Higher Secondary Certificate (HSC); Percentage: 88%</i>	Coimbatore, India Completed 2022

EXPERIENCE

Antz AI <i>AI Engineering Intern</i>	May 2025 – Jul 2025 Hybrid
- Worked on developing Retrieval-Augmented Generation (RAG) pipelines by integrating embedding models with vector databases to enable semantic search and context-aware information retrieval for LLM-based applications. - Contributed to prompt engineering, response evaluation, and retrieval optimization to improve answer quality, while testing and refining AI workflows for scalable and reliable document intelligence solutions.	

PROJECTS

Context-Aware Medical Q&A <i>LlamaIndex, Qdrant, Agno Agent</i>	Aug 2025 – Feb 2026
Developed an end-to-end Retrieval-Augmented Generation (RAG) system by implementing document ingestion, intelligent chunking, embedding generation, and semantic retrieval using LlamaIndex and Qdrant. Integrated Ollama's TinyLlama with Agno Agent to orchestrate context-aware query processing and generate domain-specific medical responses with improved retrieval accuracy.	
Cloud Pattern Analysis for Aviation Safety <i>Computer Vision, ML</i>	Aug 2024 – Nov 2024
Engineered a computer vision pipeline using Digital Image Processing techniques including image enhancement, segmentation, feature extraction, and texture analysis to classify cloud formations. Trained Machine Learning models to estimate turbulence risk and generate real-time aviation safety advisories from onboard camera feeds.	
Formula 1 Fan Hub <i>Web Development, UI/UX</i>	Jan 2024 – May 2024
Designed and developed a responsive Formula 1 web application featuring race schedules, multimedia streaming, merchandise browsing, and interactive user interfaces. Applied responsive web design principles and optimized frontend performance to enhance accessibility and user engagement across multiple devices.	
State & City Wise Crime Database <i>DBMS, SQL, Backend</i>	July 2023 – Oct 2023
Designed and implemented a normalized relational database for managing state- and city-wise crime records using SQL. Developed efficient CRUD operations, optimized relational queries, and integrated analytical visualizations to support scalable crime data management and reporting.	
Movie Ticket Booking System <i>Java, OOP</i>	Aug 2022 – Dec 2022
Developed a Java-based movie ticket reservation application by applying object-oriented design principles including encapsulation, inheritance, and polymorphism. Implemented modular booking workflows, seat management, and confirmation generation to deliver a structured and user-friendly reservation system.	

CERTIFICATIONS & RESEARCH

Professional Certifications: Google Cloud Foundations , AWS Cloud Foundations, Introduction to Ethical Hacking
Academic Foundations: Artificial Intelligence, Computer Networks, Cryptography, Cloud Computing
Research Publication: *Retrieval-grounded Multimodal Clinical Language Modeling*, accepted for publication at **ICTIS 2026** (Springer LNNS), introducing a multimodal Retrieval-Augmented Generation (RAG) framework for privacy-preserving clinical Large Language Models with improved factual reliability.

TECHNICAL SKILLS

Core Areas: AI/ML, Large Language Models, Retrieval-Augmented Generation, Computer Vision, Full-Stack Development, Prompt Engineering, Quality Assurance, Data Analytics, Database Management Systems, Cloud Computing, Backend Development
Languages & Technologies: Python, Java, C++, SQL, HTML, CSS, JavaScript, LlamaIndex, Ollama, OpenCV, MySQL, Git/GitHub, GCP, AWS, Postman, Jira, Docker, Agile, Scrum